THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 11–14: **Exploring Some Geometric Themes, Tales by Dots and Lines, Algebra Play, Area** | Class 8 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** 8	$n = 6$, faces $= n + 2$
2	**(d)** 52	$(10 \times 40 + 15 \times 60) \div 25$
3	**(a)** $5x + 12$	$3x + 12 + 2x$
4	**(d)** 252 cm²	$300 - 48$
5	**(a)** 12	Base has $n = 6$ sides; edges $= 2n$
6	**(c)** 42	Sum changes by 6
7	**(b)** $x^2 + 7x + 10$	Multiply term by term
8	**(a)** 12	$b_1h_1 = b_2h_2$: $180/15$
9	**(b)** 12	Each edge is shared by 2 faces: $E = 3F/2$
10	**(b)** 42.5	Arrange in ascending order first
11	**(a)** 15	$3 \times 9 - 5 \times 4 + 8$
12	**(b)** 216 cm²	Other half-diagonal $= \sqrt{15^2 - 9^2} = 12$; area $= \frac{1}{2} \times 18 \times 24$
13	**(a)** 20	$V = 2 + E - F$
14	**(c)** 40	Required total 180 minus 140
15	**(c)** $10x + 14$	Perimeter $= 2(l + b)$

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	12 edges (1); sum $= 4(15 + 9 + 6) = 120$ cm (1).	1+1
17	Total marks $= 15 \times 60 + 10 \times 70 = 1600$ (1); mean $= 64$ (1).	1+1
18	$2x + 10 + 3x - 3$ (1); $5x + 7$ (1).	1+1
19	$l + b = 40$, so $b = 14$ cm (1); area $= 26 \times 14 = 364$ cm² (1).	1+1
20	$n = 12$ (1); faces 14, vertices 24 (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	(i) 8 (1); (ii) $(4 - 2)^3 = 8$ (1); (iii) $64 - 8 = 56$ (1).	1+1+1
22	Total weight $= 10 \times 42 + 15 \times 48 = 1140$ kg (2); mean $= 45.6$ kg (1).	2+1
23	Perimeter $9x + 9$ cm (2); for $x = 2$: $9 \times 2 + 9 = 27$ cm (1).	2+1
24	Inner rectangle $= (40 - 4) \times (30 - 4) = 936$ m² (2); path $= 1200 - 936 = 264$ m² (1).	2+1
	OR: $E = 3F/2 = 15$ (1); $V = 2 + E - F = 7$ (2).	1+2
25	Sum changes by 10 (1); new sum $= 140$ (1); new mean $= 28$, change $= 2$ (1).	1+1+1
	OR: $x^2 + y^2 = (x + y)^2 - 2xy$ (1) $= 11^2 - 2 \times 28$ (1) $= 65$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Prism: 7, 10, 15; pyramid: 6, 6, 10 (2); differences: faces 1 (1), vertices 4 (1), edges 5 (1).	2+1+1+1
	OR: (i) changes: 4, 5, 3, -4, -5 (2); (ii) mean $= 24$ (2); (iii) largest change 5°C (1).	2+2+1
27	$x^2 + 8x + 15 - x^2 - 4x$ (2); $4x + 15$ (1); at $x = 2$: 23 (2).	2+1+2
	OR: Rectangle $= 200$ cm² (1); triangle $= \frac{1}{2} \times 20 \times 8 = 80$ cm² (2); total $= 280$ cm² (1); percentage $= 28.57\%$ (1).	1+2+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 11–14: **Exploring Some Geometric Themes, Tales by Dots and Lines, Algebra Play, Area** | Class 8 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** 156 m²	$(20+4)(15+4)-20\times15$
2	**(d)** 8	Inner cube of side 2
3	**(a)** 20	$k(n+1)/2 = 4\times10/2$
4	**(a)** 53	$(x+y)^2 = x^2+y^2+2xy$
5	**(b)** 150 cm²	Other leg $=\sqrt{25^2-15^2}=20$; area $=\frac{1}{2}\times15\times20$
6	**(c)** 48	Length multiplies by $\frac{4}{3}$ each stage: $27\times(4/3)^2$
7	**(c)** 50%	Rise $=10$; $(10/20)imes100$
8	**(a)** $6n+6$	$2n+(2n+2)+\ldots$: $6n+6$
9	**(d)** 50 cm²	Area $=d^2/2=10^2/2$
10	**(d)** 13	$1+3+\cdots+3^2=(3^3-1)/2$
11	**(b)** 41	$(20\times35+30\times45)\div50$
12	**(a)** $9x+10$	$5x+10+4x$
13	**(b)** 380 cm²	$450-70$
14	**(b)** 10	$n=8$, faces $=n+2$
15	**(a)** 36	Sum changes by 12

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	New mean $=27+6=33$ (1+1).	1+1
17	$x^2+9x+4x+36$ (1); $x^2+13x+36$ (1).	1+1
18	Half-diagonal$^2=50^2-14^2=2304$ (1); other diagonal $=96$ cm; area $=1344$ cm² (1).	1+1
19	Segments $=16$ (1); total $=112$ cm (1).	1+1
20	Numbers: 3, 6, ..., 30 (1); mean $=k(n+1)/2=16.5$ (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Area $=24\times10=240$ cm² (2); height on side 16 $=240\div16=15$ cm (1).	2+1
22	Number $=3^3=27$ (1); shaded area $=128\times(3/4)^3=54$ cm² (2).	1+2
23	Median stays $=49$ since the middle values are unchanged (1); mean increases by $152/7$ (1) to 72.71 (1).	1+1+1
24	Terms $2n, 2n+2, \ldots, 2n+4$ (1); sum $6n+6$ (1); for $n=4$ the numbers add to 30 and 30 agrees (1).	1+1+1
	OR: Hypotenuse $=15$ cm (1); area $=54$ cm² (1); altitude $=2\times54/15=7.2$ cm (1).	1+1+1
25	(i) 8 (1); (ii) $(5-2)^3=27$ (1); (iii) $125-27=98$ (1).	1+1+1
	OR: Total weight $=20\times36+30\times41=1950$ kg (2); mean $=39$ kg (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Stage 0: $3\times9=27$ cm (1); stage 1: $27\times\frac{4}{3}=36$ cm (2); stage 2: 48 cm (2).	1+2+2
	OR: Total $=20\times50+30\times60=2800$ (2); mean $=56$ (2); change $=6$ (1).	2+2+1
27	$(10a+b)+(10b+a)=11a+11b=11(a+b)$ (3); for 3 and 9: 132 (1); divisible by 11 (1).	3+1+1
	OR: $h=2A/(a+b)=10$ cm (2); parallelogram area $=18\times10=180$ cm² (2); triangle left $=70$ cm² (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 11–14: **Exploring Some Geometric Themes, Tales by Dots and Lines, Algebra Play, Area** | Class 8 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** $x^2 + 7x + 12$	Multiply term by term
2	**(c)** 15	$b_1h_1 = b_2h_2$: $240/16$
3	**(a)** 16	Base has $n = 8$ sides; edges $= 2n$
4	**(c)** 40.5	Arrange in ascending order first
5	**(b)** 40	$4 \times 12 - 6 \times 3 + 10$
6	**(d)** 480 cm²	Other half-diagonal $= \sqrt{26^2 - 10^2} = 24$; area $= \frac{1}{2} \times 20 \times 48$
7	**(b)** 18	Each edge is shared by 2 faces: $E = 3F/2$
8	**(a)** 0	Required total 150 minus 150
9	**(c)** $14x + 14$	Perimeter $= 2(l + b)$
10	**(d)** 324 m²	$(30 + 6)(18 + 6) - 30 \times 18$
11	**(a)** 60	$V = 2 + E - F$
12	**(c)** 30	$k(n + 1)/2 = 5 \times 12/2$
13	**(d)** 58	$(x + y)^2 = x^2 + y^2 + 2xy$
14	**(b)** 210 cm²	Other leg $= \sqrt{29^2 - 20^2} = 21$; area $= \frac{1}{2} \times 20 \times 21$
15	**(a)** 64	Inner cube of side 4

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	$(10a + b) - (10b + a) = 9(a - b)$ (1) $= 9 \times 2 = 18$ (1).	1+1
17	Area $= d^2/2 = \frac{18^2}{2} = 162$ m² (1); side $= \sqrt{162}$ m (1).	1+1
18	12 edges (1); sum $= 4(14 + 8 + 5) = 108$ cm (1).	1+1
19	Total marks $= 20 \times 45 + 30 \times 55 = 2550$ (1); mean $= 51$ (1).	1+1
20	$4x + 12 + 2x - 12$ (1); $6x$ (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Perimeter $10x + 12$ cm (2); for $x = 3$: $10 \times 3 + 12 = 42$ cm (1).	2+1
22	Inner rectangle $= (50 - 6) \times (36 - 6) = 1320$ m² (2); path $= 1800 - 1320 = 480$ m² (1).	2+1
23	$E = 3F/2 = 27$ (1); $V = 2 + E - F = 11$ (2).	1+2
24	Sum changes by 10 (1); new sum $= 185$ (1); new mean $= 37$, change $= 2$ (1).	1+1+1
	OR: $x^2 + y^2 = (x + y)^2 - 2xy$ (1) $= 13^2 - 2 \times 40$ (1) $= 89$ (1).	1+1+1
25	Area $= 30 \times 12 = 360$ cm² (2); height on side $18 = 360 \div 18 = 20$ cm (1).	2+1
	OR: Number $= 3^2 = 9$ (1); shaded area $= 320 \times (3/4)^2 = 180$ cm² (2).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Prism: 8, 12, 18; pyramid: 7, 7, 12 (2); differences: faces 1 (1), vertices 5 (1), edges 6 (1).	2+1+1+1
	OR: (i) changes: 5, 6, 4, -5, -6 (2); (ii) mean $= 27.5$ (2); (iii) largest change 6°C (1).	2+2+1
27	$x^2 + 9x + 14 - x^2 - 3x$ (2); $6x + 14$ (1); at $x = 4$: 38 (2).	2+1+2
	OR: Rectangle $= 288$ cm² (1); triangle $= \frac{1}{2} \times 24 \times 10 = 120$ cm² (2); total $= 408$ cm² (1); percentage $= 29.41\%$ (1).	1+2+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.